SAMPLE

Famous Astronomers: Christiaan Huygens

Light, Rings, and Pendulums

Aira Astralis

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Introduction

A few figures in the history of science stand at the hinge between two ages, and Christiaan Huygens is one of them. Born in the Netherlands in 1629, working in the decades when telescopes were new instruments still being learned, he found the largest moon of Saturn, explained the rings of Saturn, saw the first surface marking ever recorded on Mars, mapped the heart of the Orion Nebula, invented the pendulum clock, founded the mathematical theory of probability in print, proposed the wave theory of light, and wrote a book speculating, with careful arguments, about the inhabitants of other worlds. He was the finest telescope maker of his century and one of its finest mathematicians, and he did the two crafts in the same room, grinding glass by day and computing orbits and clocks by night.

The reader new to his story may know him best through the modern mission that carried his name, the probe that descended through the haze of Titan in 2005, the very moon he discovered with a homemade telescope in 1655. That pairing, the discovery from a rooftop in the Hague and the landing three and a half centuries later, is the emblem of this book: his observations were so well made that the science is still building on them, and his questions were so well asked that the answers are still arriving.

This book keeps the series order: first the finding of the man, then what he did and how, the traits that carried him, the physics inside his instruments, the craft of his observing, the inventions and features that matter, his place among the century's astronomers, what the modern science makes of him, the work he set going, the discoveries on the permanent record, the difficulties and open riddles, where his questions now stand, the small facts, and the closing measure of his importance. The account stays evergreen, holding to the settled facts of the life and the permanent physics of the questions, so that it wears well on any later shelf.

A note on the shape of his career, since it organizes everything that follows. Huygens belonged to the first generation that could take the telescope for granted, and he spent his inheritance and his genius improving it; he belonged to the first generation that could put mathematics into timekeeping, and he made the clock a precision instrument; and he belonged to the small circle of minds that debated the nature of light while Newton was still young, and he staked the losing side of that debate with reasoning that the twentieth century vindicated. The reader will meet him as a young prodigy in the Hague, a grand master of instruments in the Paris Academy, and an old man at the seaside finishing a book about the inhabitants of the planets, and the three portraits are the same man.

A word on how the book is built, before the history begins. The chapters hold the chronology loosely and the series order firmly, so a fact of the boyhood may appear among the characteristics and a fact of the last years in the science; the reader wanting the straight chronology can take the opening chapters and the closing ones in order. The sources behind the account are the standard ones: the collected works and correspondence in the great modern edition, the manuscripts preserved in the family's papers, the published books of his lifetime, and the modern studies of the Dutch golden age's science. Where the evidence thins out, as any priority race's does, the thinness is said rather than smoothed, since the era's disputes are themselves part of the evidence of how the science worked.

A paragraph on the town that made him belongs at the history's front. The Hague of his youth was the seat of a republic that ran on trade, toleration, and printing, and the town's combination, money for instruments, peace for inquiry, and presses for results, was the century's best soil for science; the reader should picture the young man walking from his father's court connections to the lens-grinders' street to the printers' row, all within the same town, and understand that his career's speed was the town's speed, a place where the whole chain of making, thinking, and publishing lay within an hour's walk of the workshop.

One more paragraph of the history belongs at its close, on the character of the fame. The century's recognition of him was immediate and complete: the astronomers adopted his Saturn solution, the clockmakers copied his governor, the mathematicians read his treatise, and the academies competed for his presence, all before he was thirty-five. The rapid fame had a cost the record also keeps: the obligation of the standard, since every claim he made after 1659 was weighed by Europe against the standard of the first claims, and the weighing explains his later caution, the delays, the anagrams, and the decades of the wave theory's gestation; the man who set the century's mark had to keep clearing it, and the book's later chapters are, in part, the story of the effort.

A final paragraph sets the tone for the book, in the man's own terms. The era's word for the largest questions was the world's system, and the man who explained Saturn's rings, timed the seconds, and reasoned about the

plurality of worlds held the whole of the system in one head; the book's chapters are the system's pieces, the glass and the geometry, the clock and the light, the sky's entries and the book of worlds, and the reader is asked at the start for the patience the pieces require, since the whole of a working life does not assemble in a paragraph. The life assembled in forty working years, in the Hague and in Paris, at the bench and at the eyepiece, and the chapters that follow are the assembly's record, piece by piece, in the order the series' readers know.

Chapter 1: The Discovery of Christiaan Huygens

The life begins in the Hague, in a great house, in the middle of the Dutch golden age.

Christiaan Huygens was born in 1629, the second son of Constantijn Huygens, a diplomat, poet, and polymath whose house was a station on the intellectual map of Europe. The father's circle included the philosophers and scientists of the age; Descartes stayed with the family and praised the boy's talent; and the education was private, broad, and in the modern languages of learning as well as the old ones. The childhood produced the era's ideal of a gentleman scholar, at home in music, mathematics, fencing, and languages, and the letters of the family show the second son already deep in geometry.

The formal studies were brief and deliberately so. He studied law and mathematics, briefly, then returned home, and the return was the making of him: the years he spent in the Hague in his twenties were spent on the questions the age had just opened, the pendulum, the colliding bodies, the grinding of lenses, and the planets. The private means of the family supported the science; the profession of the father opened the correspondence; and the arrangement, a young man with instruments, means, and the best scientific mail in Europe, produced results almost at once.

The discovery of Huygens, in the public sense, came in 1655 and 1656. In March 1655 he turned a telescope of his own making, with a lens he had ground himself, on Saturn, and found a small point of light near the planet that proved, over the nights that followed, to be a satellite circling it; in the same season he found the great nebula in Orion resolvable into stars, with a sketch that mapped what no eye had seen before. Within two years he had solved the mystery of Saturn's shape, explaining the planet's strange appendages as a thin, flat ring, nowhere touching, and announcing the solution in the manner of the age, first as an anagram and then in a book on the Saturn system.

The Saturn discovery made him famous at twenty-seven. The anagram announcing the ring, sent to observers across Europe, was the custom of the age for protecting a claim while the evidence was checked; the book that followed, a slender volume on the system of Saturn, made the case entire, with the observations, the reasoning, and the theory that explained why the ring had appeared in past decades as handles, arms, and phases. The astronomers of Europe read it, accepted it, and adopted his instruments; the reputation was made in a single stroke, and the rest of his career was, in a sense, the working out of what that spring in the Hague had begun.

The dates of the life, kept in one line as the record holds them: born in the Hague 1629; educated at home and briefly at university; the discovery years 1655 to 1659, with Titan, the Orion Nebula, the ring of Saturn, and the first markings on Mars; the pendulum clock patented in 1656 and developed into the age's standard timekeeper; the move to Paris and the new Academy of Sciences in 1666; the great book on the pendulum in 1673; the wave theory of light developed in the 1670s and published in 1690; the return to the Hague in 1681; the last years at the seaside town, working and corresponding and finishing the book on the plurality of worlds; and death in the Hague in 1695, with the plurality book in manuscript, printed by his brother three years later. The chapters that follow are the reasons for the line.

A note on the language of his working life completes the chapter. He thought in Dutch, published in Latin, administered in French, and corresponded in all of them plus the era's diplomatic formulas; the multilingual habit was the century's necessity and, in his case, a working tool, since the Netherlands' science lived on the international mails and the Paris Academy worked in a language he had to master at thirty-seven. The reader should picture the same mind moving without friction between the workshop's Dutch, the geometry's Latin, and the Academy's French, and should credit the frictionlessness as part of the craft: the century's science was a correspondence network with instruments attached, and a scientist who could not write the network's languages was, whatever his talent, working alone.

A closing word on what made him run: his energy was husbanded like an estate. He worked with the era's routines, rose to the workshop and the letters, kept the evenings for observation and the nights for the sky, and arranged his health, his travels, and his contracts so that the science ran continuously for forty years; the evidence

of the management is the output, since the notebooks do not gap for the decades of illness, war, and move that the letters record. The reader who wants the lesson in one sentence has it in the man's own working rule, kept in the papers: the instrument ready, the sky checked, and the night never wasted, a rule the reader has met before in this series and will meet again, since the sky, in every century, has belonged to whoever kept it.

A note on the medicine of his era, which the reader may wonder about. His life's health, in the record, was the era's own: recurring illness, taken seriously and managed with the century's best advice, which was rest, the seaside, and the correspondence kept up from bed; the papers show the science continuing through the seasons of convalescence, with the lens work delegated to the brother and the geometry done propped. The note belongs in the understanding because the era's medicine was poor and the man's output was constant, and the reconciliation of the two is character: the work was arranged so that no illness could stop it, the notebooks carried to the sickroom, the sky waited for, and the whole apparatus, in the deepest sense, a hospital that never closed.